

Assignment 3

(Sep. 28-Oct. 5)

Problem 1:

For each of the following linear programs determine whether it is **infeasible**, **unbounded**, or if it has an **optimal point**. In each case, justify your answer (e.g. with a certificate that the linear program has this property).

(a)

$$\begin{aligned} \min_{x_1, x_2} \quad & x_1 + 2x_2 \\ \text{s.t.} \quad & -x_1 \leq 0, \\ & -x_2 \leq 0, \\ & -x_1 - x_2 \leq -1. \end{aligned}$$

Solution: This is a linear program of the form

$$\begin{aligned} \min \quad & c^\top x \\ \text{s.t.} \quad & Ax \leq b, \end{aligned}$$

where

$$c = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \quad A = \begin{pmatrix} -1 & 0 \\ 0 & -1 \\ -1 & -1 \end{pmatrix}, \quad b = \begin{pmatrix} 0 \\ 0 \\ -1 \end{pmatrix}.$$

A feasible point is

$$x^* = \begin{pmatrix} 1 \\ 0 \end{pmatrix},$$

whose objective value is

$$c^\top x^* = 1.$$

For a certificate of optimality, take

$$y^* = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}.$$

Clearly $y^* \geq 0$, and

$$A^\top y^* = \begin{pmatrix} -1 & 0 & -1 \\ 0 & -1 & -1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ -2 \end{pmatrix} = -c.$$

Moreover,

$$-b^\top y^* = -(0 \ 0 \ -1) \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} = 1.$$

(b) Consider

$$\begin{array}{ll}\min_{x_1, x_2} & -x_1 - x_2 \\ \text{s.t.} & x_1 \leq 2, \\ & x_2 \leq 3, \\ & -x_1 \leq 0, \\ & -x_2 \leq 0.\end{array}$$

Solution: This is a linear program of the form

$$\begin{array}{ll}\min & c^\top x \\ \text{s.t.} & Ax \leq b,\end{array}$$

where

$$c = \begin{pmatrix} -1 \\ -1 \end{pmatrix}, \quad A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ -1 & 0 \\ 0 & -1 \end{pmatrix}, \quad b = \begin{pmatrix} 2 \\ 3 \\ 0 \\ 0 \end{pmatrix}.$$

Take

$$x^* = \begin{pmatrix} 2 \\ 3 \end{pmatrix}.$$

It is feasible and

$$c^\top x^* = -5.$$

For the certificate of optimality, take

$$y^* = \begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \end{pmatrix}.$$

Then $y^* \geq 0$ and

$$A^\top y^* = \begin{pmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} = -c.$$

Also,

$$-b^\top y^* = -\begin{pmatrix} 2 & 3 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \end{pmatrix} = -5.$$

Therefore $x^* = (2, 3)$ is optimal.

(c)

$$\begin{array}{ll}\min_{x_1, x_2} & -x_1 + x_2 \\ \text{s.t.} & -x_1 \leq 0, \\ & -x_2 \leq 0.\end{array}$$

Solution: This is a linear program of the form

$$\begin{array}{ll} \min & c^\top x \\ \text{s.t.} & Ax \leq b, \end{array}$$

where

$$c = \begin{pmatrix} -1 \\ 1 \end{pmatrix}, \quad A = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}, \quad b = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

The point $x^{(0)} = (0, 0)$ is feasible. Consider the direction

$$d = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

We have

$$Ad = \begin{pmatrix} -1 \\ 0 \end{pmatrix} \leq 0,$$

so d lies in the recession cone of the feasible region. Moreover,

$$c^\top d = -1 < 0.$$

Hence, for every $t \geq 0$,

$$x^{(0)} + td = (t, 0)$$

is feasible, while

$$c^\top (x^{(0)} + td) = -t \longrightarrow -\infty.$$

Therefore the linear program is unbounded below.

(d)

$$\begin{array}{ll} \min_{x_1, x_2} & x_1 + x_2 \\ \text{s.t.} & x_1 \leq 0, \\ & -x_1 \leq -1. \end{array}$$

Solution: This is a linear program of the form

$$\begin{array}{ll} \min & c^\top x \\ \text{s.t.} & Ax \leq b, \end{array}$$

where

$$c = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad A = \begin{pmatrix} 1 & 0 \\ -1 & 0 \end{pmatrix}, \quad b = \begin{pmatrix} 0 \\ -1 \end{pmatrix}.$$

A certificate for infeasibility is

$$y = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Indeed, $y \geq 0$ and

$$A^\top y = \begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

On the other hand,

$$b^\top y = \begin{pmatrix} 0 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = -1 < 0.$$

If a point x satisfied $Ax \leq b$, multiplying the inequalities by the nonnegative vector y would give

$$0 = y^\top Ax \leq y^\top b = -1,$$

a contradiction. Hence the system is infeasible.

Problem 2:

A company has a set J of jobs to be assigned to a set I of employees. Employee $i \in I$ needs c_{ij} hours to complete job $j \in J$. The company wants to complete all jobs by assigning exactly one job to each employee, while minimizing the total number of hours spent to complete all jobs. For this reason, the company has formulated the following IP:

$$\begin{aligned} \min \quad & \sum_{i \in I} \sum_{j \in J} c_{ij} x_{ij} \\ \sum_{i \in I} x_{ij} &= 1 \quad \forall j \in J \\ \sum_{j \in J} x_{ij} &= 1 \quad \forall i \in I \\ x &\geq 0, \text{ integer} \end{aligned}$$

Consider the LP obtained by the above IP after ignoring the integrality constraints, that is:

$$\begin{aligned} \min \quad & \sum_{i \in I} \sum_{j \in J} c_{ij} x_{ij} \\ \sum_{i \in I} x_{ij} &= 1 \quad \forall j \in J \\ \sum_{j \in J} x_{ij} &= 1 \quad \forall i \in I \\ x &\geq 0, \end{aligned}$$

Show that the above LP is feasible (i.e., that it admits at least one feasible solution) if and only if $|I| = |J|$.

Note: to show that the LP is feasible, you need to provide one feasible solution. To show that the LP is infeasible, provide a certificate of infeasibility as in Proposition 2.1 (p. 46) in the textbook.

Solution: First, assume that $|J| = |I|$. In this case, a feasible solution can be obtained by setting $x_{ij} = 1$ if $i = j$, and $x_{ij} = 0$ otherwise. This corresponds to assign job 1 to employee 1, job 2 to employee 2, and so on. Note that $x \geq 0$, and it trivially satisfies all equality constraints.

Now, assume that $|J| > |I|$.

Let us denote the set of equality constraints of our LP by $Ax = \mathbf{1}$. Note that A has $|J| + |I|$ rows and $|I| \cdot |J|$ columns, since each column of the matrix A is associated to a variable and we have $|I| \cdot |J|$ variables. Let $A_{(ij)}$ be the column of A associated to the variable x_{ij} . Note that $A_{(ij)}$ has *exactly* two nonzero entries: it has a 1 in position j (since variable x_{ij} appears in the constraint for job j , that is exactly the j -th constraint) and it has a 1 in position $|J| + i$ (since variable x_{ij} appears in the constraint for employee i , that is exactly the $(|J| + i)$ -th constraint).

In this case, we construct a vector y as follows: $y_k = -1$ for $k = 1, \dots, |J|$ and $y_k = 1$ for $k = |J| + 1, \dots, |J| + |I|$. Note that, for each column $A_{(ij)}$ of A we get $y^T A_{(ij)} = 0$, and therefore $y^T A = \mathbf{0}$. On the other hand, $y^T \mathbf{1} = -|J| + |I| < 0$. That is, y is a certificate of infeasibility.

Similarly, if $|J| < |I|$, we construct a vector y as follows: $y_k = 1$ for $k = 1, \dots, |J|$ and $y_k = -1$ for $k = |J| + 1, \dots, |J| + |I|$. Note that $y^T A = \mathbf{0}$ and $y^T \mathbf{1} = |J| - |I| < 0$. That is, y is a certificate of infeasibility.

Problem 3:

(a) Here is a linear program

$$\begin{array}{ll}\text{minimize} & 3x_1 + 2x_2 + x_3 \\ \text{subject to} & x_1 + x_2 + x_3 \leq 8, \\ & 2x_1 - x_2 + x_3 \geq 2, \\ & x_1 + 3x_2 - x_3 \leq 10, \\ & -x_1 + x_2 + 2x_3 \geq 1. \quad x_1, x_2, 2x_3 \geq 0.\end{array}$$

Write an equivalent linear program in standard equality form.

Solution: To put a LP in standard equality form, make the objective maximization by negating the objective. Then, introduce slack variables $s_1, \dots, s_5$ for each inequality, and use this to represent the difference between the inequality and an equality.

$$\begin{array}{ll}\text{maximize} & -3x_1 - 2x_2 - x_3 \\ \text{subject to} & x_1 + x_2 + x_3 + s_1 = 8, \\ & 2x_1 - x_2 + x_3 - s_2 = 2, \\ & x_1 + 3x_2 - x_3 + s_3 = 10, \\ & -x_1 + x_2 + 2x_3 - s_4 = 1, \\ & x_1, x_2, x_3, s_1, s_2, s_3, s_4 \geq 0.\end{array}$$

(b) Consider the following matrix (where the top row just numbers the columns).

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 0 & 1 & 2 & 1 \\ 0 & 1 & 1 & 1 & 2 \\ 0 & 0 & 0 & 1 & 1 \end{pmatrix}.$$

which of the following sets of columns are bases of this matrix?

$$\{1, 2, 5\}, \{1, 2\}, \{1, 2, 3\}, \{1, 3, 5\}$$

Solution: A basis is a set of linearly independent columns with the same number of elements as there are rows of the matrix.

Only $\{1, 2, 5\}$ and $\{1, 3, 5\}$ are bases; $\{1, 2\}$ does not have enough elements, and $\{1, 2, 3\}$ is linearly dependent.

(c) Consider the same matrix as in (b) in the equation

$$Ax = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}.$$

Put this linear equation in canonical form, using the basis $\{1, 2, 4\}$.

Solution:

$$\begin{bmatrix} 1 & 0 & 1 & 0 & -1 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix} x = \begin{pmatrix} -5 \\ -1 \\ 3 \end{pmatrix}$$